

Verification and Validation Report: Software Engineering

Team 11, technically functional

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1 Revision History

Date	Version	Notes
Date 1	1.0	Notes
Date 2	1.1	Notes

2 Symbols, Abbreviations and Acronyms

symbol	description
T	Test

[symbols, abbreviations or acronyms – you can reference the SRS tables if needed —SS]

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3 System evaluation

3.1 New user account creation

Table 1: Software Test Cases for User Account Creation

Test Case ID	user_acc_create
Control	Automatic
Rationale	The system must be able to create and store a new account
State	User does not have an account on the application.
Input	Govt id (name, date of birth), license number (PT) OR invite code (patient), email, password (new)
Output	'Account created successfully' displayed on page and user is successfully added to the user database
How the test will be performed	The test will be conducted with the use of a script which will input valid, invalid and edge case values
Test Case ID	Invalid_formatCreate
Control	Automatic
Rationale	The data input by the user has to be in the correct format for our system to accept it as valid.
State	User does not have an account on the application.
Input	Incorrect format of license number/invite code OR incorrect format of date of birth, email, password(new)
Output	System displays 'incorrect format of one or more inputs'
How the test will be performed	The test will be conducted with the use of a script which will input valid, invalid and edge case values
Test Case ID	Invalid_sign_up_creds
Control	Automatic

Rationale	Successful verification for both types of users is needed to use the application
State	User does not have an account on the application.
Input	Incorrect license number/mismatch of license number and govt id OR incorrect format of invite code, email, password(new)
Output	System displays 'License number cannot verified' OR 'invalid invite code'
How the test will be performed	The test will be conducted with the use of a script which will input valid, invalid and edge case values
Test Case ID	Acc_exists
Control	Automatic
Rationale	New users should not already have an account on the system
State	Account already exists on system
Input	Govt id (name, date of birth), license number/invite code, email, password
Output	'Account already exists. Wrong password. Try again.'
How the test will be performed	The test will be conducted with the use of a script which will input valid, invalid and edge case values

3.2 PT focused testing

Table 2: Software Test Cases for Physical Therapist (PT) Functions

Test Case ID	PT_deletes_patient_acc
Control	Automatic
Rationale	This ensures PTs control over patients' accounts. PT can remove patient(s) profiles from the application.
State	PT and patient have account on application, PT has the patient on their patient list

Input	PT requests a specific patient's account deletion
Output	Account deleted from user database and PT patient list. Patient is unable to log in ('Account not found').
How the test will be performed	The test will be conducted with the use of a script which will input valid, invalid and edge case values
Test Case ID	PT_adjusts_patient_exercise
Control	Automatic
Rationale	The system allows the PT to obtain an overview of patient performance and adjust different parameters accordingly (eg. number of repetitions).
State	PT and patient have an account on application, PT has patient on their patient list
Input	PT accesses the patient's profile page and adds/modifies exercise
Output	Patient exercise criteria is modified.
How the test will be performed	The test will be conducted with the use of a script which will input valid, invalid and edge case values
Test Case ID	PT_sends_invite
Control	Automatic
Rationale	A patient can only sign up if they have an invite code from the PT.
State	Patient does not have an account on the application while PT does.
Input	PT adds patient's information and contact method (where the code will be sent). Contact methods may include: email, text messages, and other forms of communication.
Output	An 'Invite sent' message is displayed on the screen
How the test will be performed	The test will be conducted with the use of a script which will input valid, invalid and edge case values

4 Nonfunctional Requirements Evaluation

4.1 Usability

4.2 Performance

4.3 etc.

5 Comparison to Existing Implementation

This section will not be appropriate for every project.

6 Unit Testing

7 Changes Due to Testing

[This section should highlight how feedback from the users and from the supervisor (when one exists) shaped the final product. In particular the feedback from the Rev 0 demo to the supervisor (or to potential users) should be highlighted. —SS]

8 Automated Testing

9 Trace to Requirements

10 Trace to Modules

11 Code Coverage Metrics

Appendix — Reflection

The information in this section will be used to evaluate the team members on the graduate attribute of Reflection.

The purpose of reflection questions is to give you a chance to assess your own learning and that of your group as a whole, and to find ways to improve in the future. Reflection is an important part of the learning process. Reflection is also an essential component of a successful software development process.

Reflections are most interesting and useful when they're honest, even if the stories they tell are imperfect. You will be marked based on your depth of thought and analysis, and not based on the content of the reflections themselves. Thus, for full marks we encourage you to answer openly and honestly and to avoid simply writing "what you think the evaluator wants to hear."

Please answer the following questions. Some questions can be answered on the team level, but where appropriate, each team member should write their own response:

1. What went well while writing this deliverable?
2. What pain points did you experience during this deliverable, and how did you resolve them?
3. Which parts of this document stemmed from speaking to your client(s) or a proxy (e.g. your peers)? Which ones were not, and why?
4. In what ways was the Verification and Validation (VnV) Plan different from the activities that were actually conducted for VnV? If there were differences, what changes required the modification in the plan? Why did these changes occur? Would you be able to anticipate these changes in future projects? If there weren't any differences, how was your team able to clearly predict a feasible amount of effort and the right tasks needed to build the evidence that demonstrates the required quality? (It is expected that most teams will have had to deviate from their original VnV Plan.)